

Jay Kim

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TECHNICAL SKILLS

Languages	Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks	Django REST Framework, FastAPI, Flask, React, Next.js, Node.js, Express.js, Testing (JUnit, PyTest)
Databases	PostgreSQL, MySQL, Oracle Database, MongoDB, Redis, AWS DynamoDB, Azure Cosmos DB
Cloud/DevOps	AWS EC2, AWS S3, Azure VM, Azure Blob Storage, Docker, Kubernetes, GitHub Actions, CI/CD
Data/ML	Apache Kafka, Apache Spark, Databricks, Snowflake, Azure Data Explorer, Pandas, PyTorch, Scikit-Learn
Tools	Git, GitHub, Jira, Azure DevOps, Postman, Splunk, Tableau, Alteryx

EXPERIENCE

Qcells

San Francisco, CA

Software Engineer | Data & Real-Time Intelligence

June 2026 – Present

- Developed microservices for telemetry observations, forecasts, metrics, and other energy-management data, giving downstream engineering teams standardized access to battery, inverter, and grid data without requiring direct Kusto queries
- Built an internal Python package that automated the delivery lifecycle for read APIs, including scaffolding, mock services, mapping, configuration, deployment, Kusto integration, and reusable testing, reducing development time by 50%
- Developed a reusable CRUD service for object storage, standardizing object creation, retrieval, updates, and deletion while reducing duplicated storage logic across platform services
- Automated build, testing, and deployment workflows with GitHub Actions and implemented Azure Monitor and Application Insights observability, reducing manual deployment effort and improving release reliability and incident detection

Capital One

McLean, VA

Python Developer & Data Analyst | Data Infrastructure, Analysis, and Engineering

August 2021 – August 2023

- Built a full-stack application using React and Python/Django REST Framework that surfaced KPIs and statistical stability metrics across 20+ bank models, replacing recurring ad hoc analyst requests with standardized automated reporting
- Engineered Apache Spark data pipelines that cleaned and transformed 100M+ rows of transaction data, reducing model execution time by 25% through targeted performance optimizations
- Developed internal Python tools that automated transaction pattern analysis for digital asset monitoring workflows, accelerating AML alert reviews by 40%
- Integrated the Holistic Risk Scoring Model into AML data pipelines and engineered predictive features from 3+ years of historical investigation outcomes, improving fraud alert escalation precision by 10%

Franklin Templeton

Pasadena, CA

Python Developer Intern | Quantitative Investment Operations

June 2020 – August 2020

- Developed a portfolio data management system and dashboard by integrating internal REST APIs with Snowflake to assess and track portfolio data for 200+ clients, increasing operational efficiency by 90%
- Migrated Alteryx portfolio tracking workflows to reusable Python libraries, enabling identification of underperforming assets and allocation analysis across fixed-income strategies
- Created a portfolio back-testing engine using Python and Alteryx to measure projected returns across 10+ fixed-income derivatives and ETF strategies using historical and benchmark return data
- Streamlined client cash data management by implementing an automated data ingestion framework for uploading and downloading client data; projected to save 300 work hours annually

Pacific Capital

Newport Beach, CA

Quantitative Analytics Intern | Portfolio Management

May 2019 – August 2019

- Designed reusable investment operations modules in Python that standardized Bloomberg feed parsing, computed market metrics, and automated recurring operations reporting
- Architected and deployed an Oracle SQL database with ETL pipelines to consolidate financial data for 120+ clients from raw sources, automating data retrieval and eliminating manual data entry
- Modeled tail-risk exposure using a volatility-weighted Python risk tool and regression analysis on historical market data under 10% S&P 500 decline scenarios

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science, Computer Science (Specialization in Artificial Intelligence)

Specialization GPA: 4.0/4.0 | Cumulative GPA: 3.5/4.0

Relevant Coursework: Advanced Internet Computing Systems and Applications, Software Development Process, Machine Learning, Knowledge-Based AI, Machine Learning for Trading, Security Operations and Incident Response, Health Informatics

University of Southern California

Los Angeles, CA

Bachelor of Science, Business Administration (Concentration in Data Science)